

## Problem 1

### Part a)

#### Part a) i)

Since  $X$  is  $\mathcal{G}$ - $\mathcal{B}(\mathbb{R})$ -measurable and  $(-\infty, x] \in \mathcal{B}(\mathbb{R})$  for all  $x$ , by the definition of measurability (Definition 3.1.2):

$$\{X \leq x\} = X^{-1}((-\infty, x]) \in \mathcal{G} \quad \forall x \in \mathbb{R}.$$

Since  $\mathbb{P}(B) \in \{0, 1\}, \forall B \in \mathcal{G}$ , then:

$$F_X(x) = \mathbb{P}(X \leq x) \in \{0, 1\} \quad \forall x \in \mathbb{R}.$$

Let  $S := \{x \in \mathbb{R} : F_X(x) = 1\}$ , and define:

$$c := \inf S = \inf\{x \in \mathbb{R} : F_X(x) = 1\}.$$

We have the following:

- $S$  is non-empty: since  $\lim_{x \rightarrow +\infty} F_X(x) = 1$  and  $F_X(x) \in \{0, 1\}, \forall x \in \mathbb{R}$ , then  $\exists M \in \mathbb{R} : F_X(M) = 1$ , and therefore  $M \in S$ .
- $S$  is bounded below (the infimum is finite): since  $\lim_{x \rightarrow -\infty} F_X(x) = 0$  and  $F_X(x) \in \{0, 1\}, \forall x \in \mathbb{R}$ ,  $\exists m \in \mathbb{R} : F_X(m) = 0$ . By monotonicity of  $F_X$ ,  $s > m, \forall s \in S$ , so  $m$  is a lower bound of  $S$  and  $c = \inf S \geq m > -\infty$ .

Therefore,  $c \in \mathbb{R}$ . We also have:

- Handwritten note: You have  $F_X(\infty) = 1$ , not  $\lim_{x \rightarrow \infty} F_X(x) = 1$ . The part above  $\uparrow$  doesn't work because  $X$  is  $\mathbb{R} \cup \{-\infty, \infty\}$ -valued.  $c$  could be  $\infty$  or  $-\infty$ .*
- For  $x < c$ ,  $F_X(x) = 0$ : Since  $c$  is a lower bound of  $S$  and  $x < c$ , we have  $x < c \leq s, \forall s \in S$ , which means  $x \notin S$ . Since  $F_X(x) \in \{0, 1\}$ ,  $S = \{x \in \mathbb{R} : F_X(x) = 1\}$  and  $x \notin S$ , then  $F_X(x) = 0$ .
  - For  $x > c$ ,  $F_X(x) = 1$ : Let  $\varepsilon := x - c > 0$  and, therefore,  $x = c + \varepsilon > c$ . Since  $c = \inf S$ ,  $\exists s \in S : s < c + \varepsilon = x$ . Since  $F_X(s) = 1$ ,  $F_X(x) \in \{0, 1\}$  and  $F_X$  is monotonically non-decreasing, then  $1 = F_X(s) \leq F_X(x)$ , hence  $F_X(x) = 1$ .
  - For  $x = c$ ,  $F_X(x) = 1$ : Since  $F_X$  is right-continuous, i.e.,  $F_X(c) = \lim_{x \downarrow c} F_X(x)$ , and  $F_X(x) = 1, \forall x > c$  (ii), then  $\lim_{x \downarrow c} F_X(x) = 1$ , so  $F_X(c) = 1$ .

By Lemma 1.4.2,  $\mathbb{P}(X = c) = F_X(c) - F_X(c-)$ , where  $F_X(c-) = \lim_{x \uparrow c} F_X(x) = 0$ , because  $F_X(x) = 0, \forall x < c$  (i). Since  $F_X(c) = 1$  (iii), then:

$$\mathbb{P}(X = c) = F_X(c) - F_X(c-) = 1 - 0 = 1.$$

□

#### Part a) ii)

Let  $Y_n = a_n(X_1 + \dots + X_n)$ . For any fixed  $m \in \mathbb{N}$  and for all  $m < n$ , we can decompose the sum:

$$\begin{aligned} Y_n &= a_n(X_1 + \dots + X_n) \\ &= a_n(X_1 + \dots + X_{m-1}) + a_n(X_m + \dots + X_n) \\ &= a_n \sum_{i=1}^{m-1} X_i + a_n \sum_{i=m}^n X_i \end{aligned}$$

We can rewrite the limit superior and limit inferior as:

$$Y^* := \limsup_{n \rightarrow \infty} Y_n = \limsup_{n \rightarrow \infty} \left( a_n \sum_{i=1}^{m-1} X_i + a_n \sum_{i=m}^n X_i \right)$$

$$Y_* := \liminf_{n \rightarrow \infty} Y_n = \liminf_{n \rightarrow \infty} \left( a_n \sum_{i=1}^{m-1} X_i + a_n \sum_{i=m}^n X_i \right)$$

Since  $X_1, \dots, X_{m-1}$  are real-valued random variables, for every outcome  $\omega \in \Omega$ , their finite sum  $\sum_{i=1}^{m-1} X_i(\omega)$  is a fixed finite real number. Since  $\lim_{n \rightarrow \infty} a_n = 0$ , it follows that  $\lim_{n \rightarrow \infty} a_n \sum_{i=1}^{m-1} X_i(\omega) = 0$  pointwise for all  $\omega \in \Omega$ . Therefore, the limits simplify to:

$$Y^* = \limsup_{n \rightarrow \infty} \left( a_n \sum_{i=m}^n X_i \right)$$

$$Y_* = \liminf_{n \rightarrow \infty} \left( a_n \sum_{i=m}^n X_i \right)$$

For any  $m$ , the sequence  $(a_n \sum_{i=m}^n X_i)_{n \geq m}$  depends exclusively on  $X_m, X_{m+1}, \dots$ , meaning  $Y^*$  and  $Y_*$  are measurable with respect to  $\mathcal{C}_m := \sigma\left(\bigcup_{k \geq m} \sigma(X_k)\right)$ . Because this holds for an arbitrary  $m \in \mathbb{N}$ , both  $Y^*$  and  $Y_*$  are measurable with respect to the terminal  $\sigma$ -algebra  $\mathcal{C}_\infty = \bigcap_{m=1}^\infty \mathcal{C}_m$ . ✓

Since the  $(X_n)_{n \in \mathbb{N}}$  are independent, the generated  $\sigma$ -algebras  $\mathcal{F}_n := \sigma(X_n)$  are independent (Definition 2.2.9). Hence, by Kolmogorov's Zero-One Law (Theorem 3.2.8),  $\mathbb{P}(C) \in \{0, 1\}$  for all  $C \in \mathcal{C}_\infty$ . Since  $Y_*$  and  $Y^*$  are  $\mathcal{C}_\infty$ -measurable, events such as  $\{Y^* \leq x\}$  belong to  $\mathcal{C}_\infty$  for all  $x \in \mathbb{R}$ , in particular, it implies  $\mathbb{P}(Y^* \leq x) \in \{0, 1\}$ . Because  $Y^*$  takes values in  $\overline{\mathbb{R}}$ , we must check its boundary behavior on the extended real line. For  $Y^*$ : just use a)i) from here

- If  $\mathbb{P}(Y^* \leq x) = 0$  for all  $x \in \mathbb{R}$ , then  $Y^* = +\infty$   $\mathbb{P}$ -a.s.
- If  $\mathbb{P}(Y^* \leq x) = 1$  for all  $x \in \mathbb{R}$ , then  $Y^* = -\infty$   $\mathbb{P}$ -a.s.
- In all other cases,  $\exists x_1, x_2 \in \mathbb{R}$  such that  $\mathbb{P}(Y^* \leq x_1) = 0$  and  $\mathbb{P}(Y^* \leq x_2) = 1$ . Since  $F_{Y^*}(x) := \mathbb{P}(Y^* \leq x) \in \{0, 1\}$  for all  $x \in \mathbb{R}$ , the set  $S := \{x \in \mathbb{R} : F_{Y^*}(x) = 1\}$  is non-empty (as  $x_2 \in S$ ) and bounded below (as  $x_1 \notin S$ ). By the same argument as in part a) i),  $c_+ := \inf S \in \mathbb{R}$  satisfies  $\mathbb{P}(Y^* = c_+) = 1$ , i.e.,  $Y^* = c_+$   $\mathbb{P}$ -a.s.

The same logic applies to  $Y_*$ . Therefore, there exist constants  $c_-, c_+ \in \overline{\mathbb{R}}$  such that:

$$\liminf_{n \rightarrow \infty} a_n(X_1 + \dots + X_n) = c_- \quad \mathbb{P}\text{-a.s.}$$

$$\limsup_{n \rightarrow \infty} a_n(X_1 + \dots + X_n) = c_+ \quad \mathbb{P}\text{-a.s.}$$

□

2/3

### Part b)

Let  $Z_n := \frac{1}{\log(n)} \log\left(\frac{1}{X_n}\right)$ . We will show that  $\limsup_{n \rightarrow \infty} Z_n = 1$   $\mathbb{P}$ -a.s. by establishing an upper bound and a lower bound:

- Upper Bound:

**Step 1. (Define the events and calculate probabilities):** Let  $\epsilon > 0$ , and  $A_n := \{Z_n > 1 + \epsilon\}$ :

$$\begin{aligned}
A_n &= \left\{ \frac{1}{\log(n)} \log \left( \frac{1}{X_n} \right) > 1 + \epsilon \right\} \\
&= \left\{ \log \left( \frac{1}{X_n} \right) > (1 + \epsilon) \log n \right\} \\
&= \left\{ e^{\log \left( \frac{1}{X_n} \right)} > e^{(1+\epsilon) \log(n)} \right\} \\
&= \left\{ \frac{1}{X_n} > n^{(1+\epsilon)} \right\} \quad (\text{since } n^{(1+\epsilon)} > 0) \\
&= \{X_n < n^{-(1+\epsilon)}\}
\end{aligned}$$

Let  $F_{X_n}$  be the distribution function of  $X_n$ .

Because  $X_n \sim U(0, 1)$  is an absolutely continuous distribution,  $\mathbb{P}(X_n < x) = \mathbb{P}(X_n \leq x) = F_{X_n}(x)$ . Because  $n \geq 2$  and  $\epsilon > 0$ , we have  $0 < n^{-(1+\epsilon)} < 1$ . Thus, we can evaluate the uniform CDF as  $F_{X_n}(x) = x$ :

$$\mathbb{P}(A_n) = \mathbb{P}(\{X_n < n^{-(1+\epsilon)}\}) = F_{X_n}(n^{-(1+\epsilon)}) = n^{-(1+\epsilon)} = \frac{1}{n^{1+\epsilon}}.$$

**Step 2. (Apply the First Borel-Cantelli Lemma):** Since  $1 + \epsilon > 1$ ,

$$\sum_{n=2}^{\infty} \mathbb{P}(A_n) = \sum_{n=2}^{\infty} \frac{1}{n^{1+\epsilon}} < \infty.$$

By the First Borel-Cantelli Lemma, we have:

$$\mathbb{P} \left( \limsup_{n \rightarrow \infty} A_n \right) = 0$$

**Step 3. ( $Z_n \leq 1 + \epsilon$  for all sufficiently large  $n$   $\mathbb{P}$ -a.s.):** Using De Morgan's laws and the definition of the limit superior of sets, we have:

$$\left( \limsup_{n \rightarrow \infty} A_n \right)^c = \left( \bigcap_{k=1}^{\infty} \bigcup_{m=k}^{\infty} A_m \right)^c = \bigcup_{k=1}^{\infty} \bigcap_{m=k}^{\infty} A_m^c = \liminf_{n \rightarrow \infty} A_n^c$$

Since  $\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = 0$ ,

$$\mathbb{P}((\limsup_{n \rightarrow \infty} A_n)^c) = \mathbb{P}(\liminf_{n \rightarrow \infty} A_n^c) = 1$$

By the formal definition of the limit inferior of a sequence of events,

$$\begin{aligned}
\liminf_{n \rightarrow \infty} A_n^c &= \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} A_n^c \\
&= \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} \{Z_n \leq 1 + \epsilon\}
\end{aligned}$$

For any outcome  $\omega \in \liminf_{n \rightarrow \infty} A_n^c$ ,

$$\omega \in \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} \{Z_n \leq 1 + \epsilon\} \implies \exists k \geq 1 : \forall n \geq k, Z_n(\omega) \leq 1 + \epsilon.$$

Therefore,  $\mathbb{P}(\liminf_{n \rightarrow \infty} A_n^c) = 1$  implies that, almost surely,  $Z_n \leq 1 + \epsilon$  for all sufficiently large  $n$ . ✓

**Step 4.** ( $\limsup_{n \rightarrow \infty} Z_n \leq 1$   $\mathbb{P}$ -a.s.): For each fixed  $\epsilon > 0$ , the (null) set  $N_\epsilon := \{\limsup_{n \rightarrow \infty} Z_n > 1 + \epsilon\}$  satisfies  $\mathbb{P}(N_\epsilon) = 0$  by Step 2. The set  $N := \bigcup_{k=1}^{\infty} N_{1/k}$  is a countable union of null sets, hence still a null set. On the complement  $N^c$ , we have  $\limsup_{n \rightarrow \infty} Z_n \leq 1 + \frac{1}{k}$  for every  $k \in \mathbb{N}$ , and letting  $k \rightarrow \infty$  gives  $\limsup_{n \rightarrow \infty} Z_n \leq 1$   $\mathbb{P}$ -almost surely.

• **Lower Bound:**

**Step 1. (Define the events and calculate probabilities):** Let  $\epsilon \in (0, 1)$  and define  $B_n := \{Z_n \geq 1 - \epsilon\}$ .

$$\begin{aligned} B_n &= \left\{ \frac{1}{\log(n)} \log \left( \frac{1}{X_n} \right) \geq 1 - \epsilon \right\} \\ &= \left\{ \log \left( \frac{1}{X_n} \right) \geq (1 - \epsilon) \log n \right\} \\ &= \left\{ e^{\log \left( \frac{1}{X_n} \right)} \geq e^{(1 - \epsilon) \log(n)} \right\} \\ &= \left\{ \frac{1}{X_n} \geq n^{(1 - \epsilon)} \right\} \\ &= \{X_n \leq n^{-(1 - \epsilon)}\} \end{aligned}$$

Since  $X_n \sim U(0, 1)$ , and  $\epsilon \in (0, 1)$  ensures  $0 < n^{-(1 - \epsilon)} < 1$  for  $n \geq 2$ , the probability evaluates to the uniform distribution:

$$\mathbb{P}(B_n) = \mathbb{P}(\{X_n \leq n^{-(1 - \epsilon)}\}) = n^{-(1 - \epsilon)} = \frac{1}{n^{1 - \epsilon}}. \quad \text{span style="color: red;">✓}$$

**Step 2. (Apply the Second Borel-Cantelli Lemma):** Since  $\epsilon > 0$ , we have  $1 - \epsilon < 1$ :

$$\sum_{n=2}^{\infty} \mathbb{P}(B_n) = \sum_{n=2}^{\infty} \frac{1}{n^{1 - \epsilon}} = \infty.$$

Because  $B_n \in \sigma(X_n)$  and the random variables  $(X_n)_{n \in \mathbb{N}}$  are independent, their generated  $\sigma$ -algebras  $\sigma(X_n)$  are independent (Remark 3.2.2). Consequently, the events  $(B_n)_{n \in \mathbb{N}}$  are also independent. Since the events are independent and their probabilities sum to infinity (by the Second Borel-Cantelli Lemma), we have:

$$\mathbb{P} \left( \limsup_{n \rightarrow \infty} B_n \right) = 1.$$

This means that almost surely,  $Z_n \geq 1 - \epsilon$  infinitely often. Therefore:

$$\mathbb{P} \left( \limsup_{n \rightarrow \infty} Z_n \geq 1 - \epsilon \right) = 1.$$

**Step 3.** ( $\limsup_{n \rightarrow \infty} Z_n \geq 1$   $\mathbb{P}$ -a.s.): Let  $\epsilon_k = \frac{1}{k}$  for  $k \in \mathbb{N}$  and let the events  $E_k := \{\limsup_{n \rightarrow \infty} Z_n \geq 1 - \frac{1}{k}\}$ , such that  $\mathbb{P}(E_k) = 1$  for all  $k \in \mathbb{N}$ . We have

$$\begin{aligned} \mathbb{P} \left( \bigcap_{k=1}^{\infty} E_k \right) &= 1 - \mathbb{P} \left( \left( \bigcap_{k=1}^{\infty} E_k \right)^c \right) \\ &= 1 - \mathbb{P} \left( \bigcup_{k=1}^{\infty} E_k^c \right) \end{aligned} \quad (1)$$

By subadditivity of probability and since  $\mathbb{P}(E_k^c) = 0$  for all  $k$ ,

$$\begin{aligned} \mathbb{P}\left(\bigcup_{k=1}^{\infty} E_k^c\right) &\leq \sum_{k=1}^{\infty} \mathbb{P}(E_k^c) = 0 \\ \implies \mathbb{P}\left(\bigcup_{k=1}^{\infty} E_k^c\right) &= 0 \quad (\mathbb{P}(\cdot) \geq 0) \\ \stackrel{(1)}{\implies} \mathbb{P}\left(\bigcap_{k=1}^{\infty} E_k\right) &= 1 \end{aligned}$$

For any outcome  $\omega \in \bigcap_{k=1}^{\infty} E_k$ , we have  $\limsup_{n \rightarrow \infty} Z_n(\omega) \geq 1 - \frac{1}{k}$  for all  $k \geq 1$ . Taking the limit as  $k \rightarrow \infty$  gives  $\limsup_{n \rightarrow \infty} Z_n(\omega) \geq 1$ . Therefore,  $\limsup_{n \rightarrow \infty} Z_n \geq 1$   $\mathbb{P}$ -almost surely. ✓

Combining the lower and upper bounds,

$$\limsup_{n \rightarrow \infty} \frac{1}{\log(n)} \log\left(\frac{1}{X_n}\right) = 1 \quad \mathbb{P}\text{-a.s.}$$

□

3/3

5/6